

KRIVA PALANKA, North Macedonia

WMO: 134930

Lat: 42.200N

Lon: 22.333E

Elev: 696

StdP: 93.24

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.7	-8.3	-15.2	1.1	-9.6	-12.3	1.4	-6.5	7.9	5.0	7.0	1.6	4.1	50	0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.3	32.9	19.8	31.2	19.1	29.6	18.6	20.7	30.4	20.0	29.2	19.2	27.9	3.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.7	13.8	24.1	16.8	13.1	23.1	16.1	12.4	22.4	63.0	30.7	60.2	29.2	57.6	28.0	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.0	6.3	5.7	-15.5	35.0	3.3	2.2	-17.8	36.6	-19.8	37.9	-21.6	39.1	-24.0	40.7		
(5)			-15.6	22.4	2.9	0.8	-17.7	23.0	-19.4	23.4	-21.0	23.8	-23.2	24.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.4	0.5	2.9	6.6	10.9	15.0	18.6	21.0	21.6	16.9	12.0	7.5	2.2
	DBStd	8.04	4.57	4.42	3.74	3.52	3.17	3.23	2.85	2.58	3.57	3.99	4.01	4.29
	HDD10.0	1008	295	200	115	29	2	0	0	0	1	27	94	244
	HDD18.3	2822	553	432	364	224	109	36	8	4	67	197	326	501
	CDD10.0	1502	0	1	9	55	158	259	342	359	209	91	18	2
	CDD18.3	275	0	0	0	0	7	44	92	106	25	1	0	0
	CDH23.3	3313	0	0	0	14	122	549	1107	1177	316	28	1	0
	CDH26.7	1182	0	0	0	1	19	157	440	472	91	3	0	0
Wind	WSAvg	3.4	3.1	3.5	3.5	3.5	3.3	3.2	3.4	3.5	3.4	3.4	3.2	3.1
Precipitation	PrecAvg	603	38	35	44	58	61	61	45	45	50	59	53	53
	PrecMax	953	107	104	105	155	125	128	138	129	145	133	97	128
	PrecMin	338	2	3	1	6	25	12	1	8	11	3	11	0
	PrecStd	131	27	22	24	34	25	30	28	29	35	39	24	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.4	17.1	20.5	25.2	29.0	32.7	35.2	35.2	31.9	26.2	20.6	15.1
		MCWB	7.9	9.3	11.2	14.1	17.3	20.0	20.5	20.2	18.4	16.3	12.9	10.4
	2%	DB	10.9	14.2	18.1	22.3	26.4	30.4	33.0	33.0	29.3	23.7	17.9	12.3
		MCWB	6.6	8.0	9.8	13.2	16.4	19.3	19.9	19.8	17.6	14.8	12.0	8.3
	5%	DB	8.9	12.0	16.0	20.1	24.5	28.6	31.1	31.2	27.1	21.7	16.0	10.1
		MCWB	5.6	7.1	8.8	12.1	15.7	18.5	19.3	19.0	16.8	14.0	11.0	7.2
	10%	DB	6.8	9.8	13.7	18.1	22.7	26.8	29.3	29.4	24.7	19.4	13.9	8.3
		MCWB	4.5	5.9	7.6	11.0	14.9	17.8	18.9	18.4	15.8	13.0	10.0	6.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.5	10.1	11.7	15.1	18.6	21.4	22.0	21.8	19.5	17.0	14.1	10.7
		MCDB	12.7	15.7	19.5	22.3	26.7	29.4	31.8	32.0	29.0	25.2	18.9	14.0
	2%	WB	7.1	8.6	10.3	13.7	17.2	20.1	20.8	20.7	18.2	15.6	12.6	8.8
		MCDB	10.3	13.4	16.8	21.0	24.9	28.7	30.4	30.9	27.3	22.0	16.9	11.6
	5%	WB	5.9	7.4	9.1	12.7	16.2	19.2	20.0	19.8	17.4	14.5	11.4	7.5
		MCDB	8.4	11.3	14.8	19.3	23.3	27.1	29.0	29.5	25.5	19.9	15.3	9.6
	10%	WB	4.7	6.1	8.1	11.5	15.3	18.3	19.2	18.9	16.4	13.5	10.3	6.2
		MCDB	6.7	9.6	13.1	16.9	21.6	25.6	27.9	27.6	23.2	18.5	13.5	7.9
Mean Daily Temperature Range	5% DB	MDBR	7.9	9.6	10.9	11.1	12.0	13.0	14.1	14.3	12.3	11.3	9.3	7.8
		MCDBR	10.6	13.0	15.0	14.8	15.4	16.0	17.2	17.1	16.4	15.0	12.2	9.6
		MCWBR	6.8	7.6	8.0	7.2	6.8	6.5	6.4	6.3	6.6	7.4	6.9	6.1
	5% WB	MCDBR	9.4	12.0	13.6	13.6	14.3	15.0	15.5	15.5	14.3	13.2	11.0	8.2
		MCWBR	6.3	7.3	7.5	6.8	6.7	6.6	6.3	6.2	6.2	6.8	6.4	5.5
Clear-Sky Solar Irradiance	taub	0.318	0.345	0.388	0.446	0.436	0.421	0.421	0.437	0.412	0.376	0.345	0.319	
	taud	2.375	2.297	2.209	2.059	2.155	2.238	2.237	2.191	2.239	2.328	2.376	2.384	
	Ebn at Noon	819	849	849	823	842	854	850	822	815	802	776	781	
	Edh at Noon	82	103	127	159	149	137	136	138	122	99	82	75	
All-Sky Solar Radiation	RadAvg	1.62	2.51	3.54	4.50	5.46	6.29	6.53	5.83	4.27	3.01	1.93	1.40	
	RadStd	0.25	0.33	0.42	0.54	0.42	0.51	0.44	0.45	0.35	0.37	0.20	0.23	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)	+0.51	N/A	N/A	N/A	N/A	N/A	N/A	-151	N/A	N/A

Nomenclature: See separate page